

# Bayesian Prior Justification Report

## Clinical Trial

Biostatistics

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## 1 Executive Summary

This report documents the Bayesian prior specification for study **Clinical Trial**, prepared by **Biostatistics** (Sponsor) and dated 07 May 2026.

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## 2 Trial Information

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| Field            | Value          |
|------------------|----------------|
| Trial / Protocol | Clinical Trial |
| Sponsor          | Sponsor        |
| Statistician     | Biostatistics  |
| Report date      | 07 May 2026    |

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## 3 Prior Specification

### 3.1 Elicitation Methodology

### 3.2 Fitted Parameters

### 3.3 Prior Summary Statistics

### 3.4 Prior Density

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## 4 Prior-Data Conflict Assessment

**Warning:** No prior-data conflict diagnostics were computed for this report.

### 4.1 Prior - Likelihood - Posterior Overlay

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## 5 Sensitivity Analysis

**Warning:** No sensitivity analysis was conducted for this report.

### 5.1 Influence of Prior Hyperparameters

### 5.2 Tornado Plot

### 5.3 Hyperparameter Influence Heatmap

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## 6 Robust and Sensitivity Priors

**Warning:** No robust, sceptical, or power priors were computed for this report.

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## 7 Regulatory Compliance Checklist

Table 1: FDA/EMA regulatory compliance checklist

| Requirement                                  | Status   |
|----------------------------------------------|----------|
| Prior elicitation method documented          | Missing  |
| Distribution family and parameters specified | Missing  |
| Expert / source of prior identified          | Missing  |
| Prior density plot provided                  | Missing  |
| Prior-data conflict assessed                 | Not run  |
| Conflict diagnostic statistics reported      | Not run  |
| Sensitivity analysis performed               | Not run  |
| Sensitivity visualisations provided          | Not run  |
| Alternative priors considered                | Not run  |
| Robust / sceptical prior computed            | Not run  |
| Regulatory report generated                  | Complete |

**Note:** This checklist maps to requirements in the FDA Draft Guidance on Bayesian Statistical Methods (2026) and the EMA Reflection Paper on the Use of Bayesian Statistics in the Evaluation of Medicinal Products.

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## 8 Appendix: Session Information

| Item               | Value               |
|--------------------|---------------------|
| R version          | 4.6.0               |
| bayprior version   | 0.1.0               |
| quarto R package   | 1.5.1               |
| Quarto CLI version | 1.9.37              |
| Platform           | x86_64-pc-linux-gnu |
| Date               | 07 May 2026         |

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*Report generated by the **bayprior** R package using `prior_report()`. Methods follow O'Hagan et al. (2006), Box (1980), Schmidli et al. (2014), Ibrahim & Chen (2000), and the FDA Draft Guidance on Bayesian Methods in Clinical Trials (2026).*